

BOND VALUATION CASE BOOKLET

STUDENT EDITION • QUESTIONS ONLY

57 Cases for Advance Reading and In-Class Discussion

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Preface

This booklet is the student edition of the Bond Valuation Case Booklet — the applied companion to Bond Valuation and Fixed-Income Analysis.

The fifty-seven cases are organized into six sessions that mirror the structure of the companion volume: foundations; yield measures; term structure and spreads; duration, convexity, and embedded options; and advanced spread analysis. Each session opens with a short orientation before its cases begin.

Every case gives you a Learning Objective, a Case Situation with all the data and assumptions you need, and a Required section listing the specific tasks to complete. Nothing beyond this is provided — the analysis, computation, and conclusion are yours to work out.

How to Use This Booklet

Each case is self-contained and can be attempted independently, but the six sessions are best worked in sequence on a first pass, since later cases assume familiarity with measures introduced earlier — particularly the progression from yield (Session 2) through duration (Session 4) to convexity and embedded options (Session 5).

Read the Case Situation carefully before attempting the Required tasks. Every number and assumption you need to solve the case is given there — no outside data is required.

All figures in this booklet are stated in Indian rupees (₹), with larger sums expressed in lakh and crore rather than in millions. Every company name is fictional; any resemblance to an actual company is unintentional.

This booklet is intended for advanced reading before class and for in-class case discussion. Bring your work attempt at each Required task to class — the discussion is far more useful once you have already committed to an answer.

Session 1: Foundations of Bond Valuation

Cases 1-5

This opening session applies the Present Value Model directly: pricing bonds at par, at a premium, and at a discount, reading the Price–Yield relationship from real numbers, and estimating a bond's price at a future date before maturity. Every later session in this casebook builds on the mechanics established here.

Case 1: Bond Valuation under the Present Value Model

VARDHMAN STEEL CORPORATION LTD

Learning Objective: To apply the Present Value Model to compute a bond's fair value from its coupon, maturity, and required yield, and to confirm the special boundary case where a bond prices at par.

Case Situation

Vardhman Steel Corporation Ltd, a mid-sized steel producer, has issued a 6-year bond with a face value of ₹1,000 and an annual coupon rate of 9%. A fixed-income analyst at a mutual fund is reviewing the bond for a new debt scheme and needs to establish its fair value. The prevailing required rate of return for bonds of comparable credit quality and maturity is currently 9% per annum. The analyst wants to confirm the bond's fair price before recommending it for inclusion in the scheme's portfolio.

Required

1. Compute the fair value of the bond under the Present Value Model.
2. State whether the bond is priced at par, premium, or discount, and explain why.

Case 2: Bond Pricing at a Premium

GANGA CEMENT INDUSTRIES LTD

Learning Objective: To value a bond whose coupon rate exceeds the market's required yield, and to interpret the resulting premium price.

Case Situation

Ganga Cement Industries Ltd issued a 7-year bond three years ago carrying an annual coupon rate of 11%, when interest rates were considerably higher. Market interest rates for comparable-quality bonds have since fallen, and the required yield on bonds of this credit quality and remaining maturity is now 8%. A wealth manager needs to determine the bond's current fair value to advise a client who is considering purchasing it in the secondary market.

Required

1. Compute the fair value of the bond given the current required yield.
2. Explain why the bond is trading above its face value.
3. State what would happen to the price if required yield rose back above the coupon rate.

Case 3: Bond Pricing at a Discount

RAJDHANI AUTO COMPONENTS LTD

Learning Objective: To value a bond whose coupon rate is below the market's required yield, and to interpret the resulting discount price.

Case Situation

Rajdhani Auto Components Ltd issued an 8-year bond carrying an annual coupon rate of 7% at a time when interest rates were relatively low. Since then, market interest rates have risen, and the required yield on bonds of comparable credit quality and remaining maturity is now 10%. A debt fund manager is evaluating whether to add this bond to the fund at its current market price and first needs to establish its fair value.

Required

1. Compute the fair value of the bond given the current required yield.
2. Explain why the bond is trading below its face value.

Case 4: The Price–Yield Relationship**BHARAT URJA POWER LTD**

Learning Objective: To illustrate, through direct computation, how a bond's price responds to changes in the required yield, and to confirm the inverse and convex nature of this relationship.

Case Situation

Bharat Urja Power Ltd has a 10-year bond outstanding with a face value of ₹1,000 and an annual coupon rate of 9%. The company's treasury team wants to understand how sensitive the bond's market value is to shifts in prevailing interest rates before deciding on the timing of a related refinancing. They have asked an analyst to compute the bond's price at a range of plausible required yields and summarize the pattern observed.

Required

1. Compute the bond's price at required yields of 7%, 9%, and 11%.
2. State whether the bond prices at a premium, at par, or at a discount in each case.
3. Describe the shape of the relationship between price and yield across the three points.

Case 5: Estimating a Bond's Future (Horizon) Price**SUVIDHA CONSUMER PRODUCTS LTD**

Learning Objective: To estimate a bond's expected selling price at a future date, prior to maturity, given an assumed required yield prevailing at that future date.

Case Situation

Suvidha Consumer Products Ltd has a 12-year bond outstanding with a face value of ₹1,000 and an annual coupon rate of 10%. An investor purchased the bond at issuance and now, four years later, wants to know what price it should fetch if sold today. Market analysts currently expect that bonds of this remaining maturity and credit quality are being priced at a required yield of 9%. The investor's advisor has been asked to estimate the bond's fair selling price.

Required

1. Determine the bond's remaining maturity at the point of sale.
2. Estimate the bond's expected selling price using the prevailing required yield.

Session 2: Yield Measures & Trading Conventions

Cases 6-16

This session works through every standard yield measure — Nominal Yield, Current Yield, Yield to Maturity, Yield to Call, and Realized Yield — followed by the market conventions of accrued interest and basis-point sensitivity that govern how bonds actually trade day to day.

Case 6: Nominal Yield

SINDHU TELECOM NETWORKS LTD

Learning Objective: To distinguish Nominal Yield from other yield measures and compute it directly from a bond's stated coupon rate.

Case Situation

Sindhu Telecom Networks Ltd has issued a bond with a face value of ₹1,000 carrying an annual coupon of ₹95. A junior analyst preparing a fact sheet for the fund's investor newsletter has been asked to report the bond's Nominal Yield alongside its current trading price of ₹1,080, and needs to confirm the correct measure and its interpretation before the newsletter is finalized, since the fund's compliance team has flagged similar disclosures for imprecise wording in the past.

Required

1. Compute the bond's Nominal Yield.
2. Explain why this figure does not change even though the bond is trading at ₹1,080.

Case 7: Current Yield

MALWA AUTO COMPONENTS LTD

Learning Objective: To compute Current Yield from a bond's coupon and market price, and to distinguish it from both Nominal Yield and Yield to Maturity.

Case Situation

An income-focused investor is comparing bonds purely on the basis of annual cash income relative to what they would pay today. One candidate is a Malwa Auto Components Ltd bond with an 8.5% annual coupon on a ₹1,000 face value, currently trading at ₹950 in the secondary market. The investor's advisor has been asked to compute the yield measure most relevant to this income-comparison objective, ahead of a broader review of similarly priced bonds across the investor's watchlist.

Required

1. Compute the bond's Current Yield.
2. Explain why Current Yield differs from the bond's 8.5% Nominal Yield.

Case 8: Yield to Maturity of a Coupon Bond

KIRTI CONSUMER PRODUCTS LTD

Learning Objective: To estimate the total return an investor earns by holding a coupon-paying bond to maturity, incorporating both coupon income and capital gain or loss.

Case Situation

A portfolio manager is reviewing a Kirti Consumer Products Ltd bond with a face value of ₹1,000, an 8% annual coupon, and 10 years remaining to maturity, currently trading at ₹920 in the market. Before deciding whether the bond meets the fund's minimum return threshold of 9%, the manager needs the single yield measure that captures the bond's complete return if held to maturity, rather than a partial measure based on coupon income alone.

Required

1. Estimate the bond's Yield to Maturity (YTM).
2. State whether the bond clears the fund's 9% minimum return threshold.

Case 9: Yield to Maturity of a Zero-Coupon Bond

HIMALAYA PHARMACEUTICALS LTD

Learning Objective: To compute Yield to Maturity for a zero-coupon bond directly, without trial and error, exploiting the absence of periodic coupons.

Case Situation

Himalaya Pharmaceuticals Ltd issued a 10-year zero-coupon bond with a face value of ₹1,000, redeemable only at maturity with no periodic interest payments. The bond currently trades at ₹456 in the market. A CFA candidate studying for the fixed-income section of the exam wants to confirm how the Yield to Maturity of a zero-coupon bond can be found directly, rather than through the iterative trial-and-error process required for coupon-paying bonds.

Required

1. Compute the bond's Yield to Maturity directly, without trial and error.
2. Explain why a closed-form solution is possible here but not for a coupon bond.

Case 10: Yield to Call

SWIFT LOGISTICS & WAREHOUSING LTD

Learning Objective: To compute Yield to Call for a callable bond and understand how it differs mechanically from Yield to Maturity.

Case Situation

Swift Logistics & Warehousing Ltd issued a callable bond with a face value of ₹1,000, an 11% annual coupon, and a call provision permitting redemption at ₹1,040 after 4 years. The bond currently trades at ₹1,075. Given the bond's high coupon relative to current market rates, an analyst believes the issuer is likely to call the bond at the earliest opportunity, and has been asked to compute the relevant yield measure under that assumption.

Required

1. Compute the bond's Yield to Call, assuming redemption at the first call date.

2. Explain what cash flows are used in this calculation instead of the cash flows used for Yield to Maturity.

Case 11: The Crossover Price and the YTM-versus-YTC Decision

COASTAL SHIPPING CORPORATION LTD

Learning Objective: To compare Yield to Maturity and Yield to Call for the same premium bond, and to determine which measure an investor should treat as the more meaningful expected return.

Case Situation

Coastal Shipping Corporation Ltd issued a callable bond with a face value of ₹1,000, a 12% annual coupon, 12 years to final maturity, and a call provision allowing redemption at ₹1,060 after 3 years. The bond now trades at ₹1,150, well above both its face value and its call price. An investment committee member wants to know which yield figure — YTM or YTC — should be used to set return expectations before approving a purchase.

Required

1. Compute the bond's Yield to Maturity.
2. Compute the bond's Yield to Call.
3. Determine which measure is more relevant for return expectations, and explain why.

Case 12: Realized Yield with Reinvestment at the Coupon Rate

SETU HOUSING FINANCE CORP LTD

Learning Objective: To compute Realized (Horizon) Yield for a bond sold before maturity, when coupons are reinvested at a rate equal to the bond's own coupon rate.

Case Situation

An investor purchased a Setu Housing Finance Corp Ltd bond three years ago at ₹950, carrying a 9% annual coupon (₹90 per year) on a ₹1,000 face value. The investor reinvested each coupon as received at a rate matching the bond's own coupon rate, and has now sold the bond for ₹980. The investor's financial planner has been asked to compute the actual compound annual return earned over the full three-year holding period.

Required

1. Compute the future value of the reinvested coupons at the end of the 3-year holding period.
2. Compute the Realized (Horizon) Yield for the full holding period.

Case 13: Realized Yield with a Different Reinvestment Rate

TRIDENT OIL & GAS LTD

Learning Objective: To compute Realized (Horizon) Yield when the actual reinvestment rate available to the investor differs from the bond's own coupon rate.

Case Situation

An investor bought a Trident Oil & Gas Ltd bond at its ₹1,000 face value, carrying a 10% annual coupon (₹100 per year). Over the three-year holding period, market interest rates fell, and the investor could only reinvest each coupon as received at 7% rather than at the bond's 10% coupon rate. The bond was sold at the end of

year three for ₹1,020. The investor wants to know the true compound annual return actually earned, given this reinvestment shortfall.

Required

1. Compute the future value of the reinvested coupons at the actual 7% reinvestment rate.
2. Compute the Realized (Horizon) Yield and compare it with the bond's 10% coupon rate.

Case 14: Realized Yield with an Estimated Future Selling Price

SUNRISE RENEWABLE ENERGY LTD

Learning Objective: To estimate Realized Yield over a holding period shorter than a bond's remaining maturity, when the future selling price itself must first be estimated from an assumed future required yield.

Case Situation

An investor purchased a 15-year, ₹1,000 face value SunRise Renewable Energy Ltd bond at issuance, carrying a 12% annual coupon, and plans to sell it after holding it for 5 years. Coupons received will be reinvested at an estimated 11% rate. The investor's advisor projects that, five years from now, bonds of the ten years' remaining maturity that SunRise's bond will then carry will be priced at a required yield of 10%. The advisor has been asked to estimate the investor's likely Realized Yield under these assumptions.

Required

1. Estimate the bond's expected selling price at the end of year 5, using the projected 10% future required yield.
2. Compute the future value of the reinvested coupons over the 5-year holding period.
3. Compute the resulting estimated Realized (Horizon) Yield.

Case 15: Accrued Interest and the Clean-versus-Dirty Price

ANAND CHEMICALS & FERTILIZERS LTD

Learning Objective: To compute accrued interest for a bond traded between coupon dates, and to distinguish the quoted (clean) price from the total amount actually paid (dirty price).

Case Situation

A bond dealer is settling a trade in an Anand Chemicals & Fertilizers Ltd bond that pays a 10% annual coupon semi-annually, with a face value of ₹1,000. The bond's quoted (clean) price is ₹1,010. Settlement falls 73 days after the last coupon payment, within a 182-day semi-annual coupon period. The back-office team needs to determine the total amount the buyer must actually pay to settle the trade, before the settlement instruction is released to the custodian.

Required

1. Compute the accrued interest owed to the seller.
2. Compute the total (dirty) price the buyer must pay to settle the trade.

Case 16: Basis-Point Sensitivity Across Different Yield Levels

ANDHRA TELECOM NETWORKS LTD

Learning Objective: To demonstrate that an identical percentage change in interest rates translates into different absolute basis-point changes depending on the prevailing starting yield level.

Case Situation

A risk analyst at Andhra Telecom Networks Ltd's treasury desk is preparing a training note for new recruits on interpreting interest rate movements. To illustrate a frequently misunderstood point, the analyst wants to compare two scenarios: a 25% relative increase applied to a low starting interest rate of 4%, and the same 25% relative increase applied to a higher starting rate of 12%, and to show recruits why traders describe rate changes in basis points rather than in percentage terms.

Required

1. Compute the new interest rate and the basis-point change when a 4% rate rises by 25%.
2. Compute the new interest rate and the basis-point change when a 12% rate rises by 25%.
3. Explain why market participants prefer basis points to describe such changes.

Session 3: Term Structure, Yield Curves & Yield Spreads

Cases 17-26

This session moves beyond a single discount rate: bootstrapping the spot rate curve, deriving forward rates, applying the Expectations and Liquidity Preference Hypotheses, and reading yield spreads for relative value — the foundation for the advanced spread measures in Session 6.

Case 17: Bootstrapping the Theoretical Spot Rate Curve

RASHTRIYA INFRA DEVELOPMENT CORP

Learning Objective: To extract an unobservable spot rate from an observed bond price and previously bootstrapped spot rates, extending the theoretical spot rate curve one maturity further.

Case Situation

A government debt analyst is building a theoretical spot rate curve for Rashtriya Infra Development Corp's guaranteed bond series. The 1-year and 2-year spot rates have already been bootstrapped from shorter-maturity instruments at 6.0% and 7.0% respectively. A 3-year, 8% annual coupon bond from the same issuer, with a face value of ₹1,000, is currently trading at ₹1,020. The analyst needs the 3-year spot rate to extend the curve, but it cannot be observed directly from any single traded instrument.

Required

1. Compute the present value of the bond's Year-1 and Year-2 cash flows using the known spot rates.
2. Solve for the 3-year spot rate implied by the bond's remaining price after removing these two present values.

Case 18: Bond Valuation Using the Spot Rate Curve

GODAVARI CHEMICALS LTD

Learning Objective: To value a coupon bond using a full spot rate curve rather than a single required yield, and to see why this approach is theoretically more accurate.

Case Situation

Godavari Chemicals Ltd has issued a 3-year bond with a face value of ₹1,000 and a 9% annual coupon. Rather than using a single discount rate for the entire cash flow stream, a fixed-income analyst wants to apply the theoretical spot rate curve currently prevailing in the market: 6.0% for one year, 6.5% for two years, and 7.0% for three years. The analyst has been asked to determine the bond's arbitrage-free fair value using this spot-rate approach and to compare it against a single-rate approximation.

Required

1. Compute the present value of each of the bond's three annual cash flows using its own maturity-matched spot rate.
2. Sum these present values to determine the bond's fair value.

3. Explain why this spot-rate value is considered more theoretically accurate than discounting all cash flows at one single rate.

Case 19: Deriving a Forward Rate from Two Spot Rates

EMERALD ENERGY HOLDINGS LTD

Learning Objective: To compute the market's implied forward interest rate for a future single-year period, using two points on the current spot rate curve.

Case Situation

A treasury analyst at Emerald Energy Holdings Ltd is planning a one-year borrowing that will begin exactly one year from now and needs an unbiased estimate of what the one-year interest rate is likely to be at that future date. The current spot rate curve shows a 1-year rate of 6.0% and a 2-year rate of 7.5%. Rather than guess, the analyst wants to extract the rate the market is already implicitly forecasting for that future one-year period, embedded in today's spot rate curve.

Required

1. Compute the forward rate for the one-year period beginning one year from now, implied by the current spot rate curve.
2. Explain what this forward rate represents about the market's expectations.

Case 20: Deriving a Multi-Year Forward Rate from the Spot Curve

BLUEWATER PORTS & LOGISTICS LTD

Learning Objective: To compute an implied forward rate spanning more than one year, extracted from two longer-maturity points on the spot rate curve.

Case Situation

Bluewater Ports & Logistics Ltd is evaluating a major capital project that will require debt financing for a three-year period beginning two years from now, once an existing facility's current financing matures. The company's treasury team wants an unbiased market estimate of borrowing costs for that future three-year window. The current spot rate curve shows a 2-year spot rate of 7.0% and a 5-year spot rate of 8.5%. The team has asked an analyst to extract the implied forward rate for the relevant three-year period.

Required

1. Compute the forward rate covering the three-year period beginning two years from now.
2. Explain how this differs mechanically from computing a single-year forward rate.

Case 21: Implied Future Short-Term Rate under the Expectations Hypothesis

VIKRAM STEEL ROLLING MILLS LTD

Learning Objective: To extract the market's implied expectation of a future one-year interest rate from the current term structure, applying the pure Expectations Hypothesis.

Case Situation

An economist advising Vikram Steel Rolling Mills Ltd's treasury committee wants to gauge what the bond market currently expects short-term interest rates to be one year from now, since this view will influence the timing of a planned working-capital facility renewal. The current 1-year spot rate is 6.0% and the current 2-year spot rate is 6.8%. Under the pure Expectations Hypothesis, the economist has been asked to extract the market's implied expectation for the one-year rate prevailing one year from today.

Required

1. Compute the market's implied expected one-year interest rate for next year, using the Expectations Hypothesis.
2. State the key assumption this calculation relies upon.

Case 22: The n-Year Rate under the Expectations Hypothesis

RAJPUTANA TEXTILES & APPAREL LTD

Learning Objective: To construct a multi-year spot rate from a full sequence of expected future one-year rates, applying the pure Expectations Hypothesis in its general, multi-period form.

Case Situation

Rajputana Textiles & Apparel Ltd's corporate treasury has obtained an economist's forecast of expected one-year interest rates over each of the next four years: 5.0% in Year 1, 6.0% in Year 2, 6.5% in Year 3, and 7.0% in Year 4. Under the pure Expectations Hypothesis, the treasury wants to know what four-year spot rate this sequence of expectations implies today, in order to benchmark a planned four-year bond issuance against the theoretically consistent market rate.

Required

1. Compute the four-year spot rate implied by the given sequence of expected one-year rates.
2. Explain the compounding logic connecting the four one-year rates to this single multi-year rate.

Case 23: The Liquidity Premium under the Liquidity Preference Hypothesis

ARYAN AUTOMOBILES LTD

Learning Objective: To extract an implied liquidity premium from the gap between an observed long-term spot rate and the pure expectations-based rate, applying the Liquidity Preference Hypothesis.

Case Situation

An analyst reviewing Aryan Automobiles Ltd's financing costs has obtained economist forecasts of expected future one-year rates: 6.0% today, 6.4% expected next year, and 6.8% expected the year after. However, the actual observed 3-year spot rate in the market is 7.5% - higher than the pure Expectations Hypothesis alone would predict from these forecasts. The Liquidity Preference Hypothesis attributes this gap to a liquidity premium investors demand for holding longer-maturity bonds. The analyst has been asked to estimate the implied premium.

Required

1. Compute the 3-year spot rate that would be implied by the pure Expectations Hypothesis alone, using the given expected rates.
2. Estimate the approximate liquidity premium embedded in the actual observed 7.5% spot rate.

Case 24: Yield Spread and Mean Reversion

DECCAN SUGAR MILLS LTD

Learning Objective: To apply the principle of mean reversion in evaluating whether a currently observed yield spread represents a genuine relative-value trading opportunity.

Case Situation

A credit analyst tracks the yield spread between Deccan Sugar Mills Ltd's corporate bonds and comparable-maturity government securities. Over the past several years, this spread has typically hovered around 80 basis points, reflecting the normal, long-run equilibrium compensation investors demand for Deccan Sugar's credit risk relative to a risk-free benchmark. This month, following a temporary sector-wide sentiment shift, the spread has widened sharply to 140 basis points. A bond desk trader wants to know whether this movement represents a genuine trading opportunity.

Required

1. Identify whether the current 140-basis-point spread should be classified as a normal or an abnormal yield spread.
2. State the trading implication that follows from the mean reversion principle.

Case 25: Comparing Quality (Credit) Spreads Across Two Issuers

AMRIT DAIRY & FOODS LTD

Learning Objective: To compute and interpret the quality (credit) spread between two bonds of differing credit quality but comparable maturity, as a measure of the market's pricing of relative default risk.

Case Situation

A credit strategist is comparing two 5-year corporate bonds of similar maturity and structure but different credit ratings: a higher-rated bond issued by Amrit Dairy & Foods Ltd, currently yielding 7.45%, and a lower-rated bond from a comparable-sized peer in the same sector, currently yielding 8.75%. The strategist has been asked to quantify the credit spread the market is demanding between these two issuers and to comment on what this spread reveals about relative perceived default risk.

Required

1. Compute the quality (credit) spread between the two bonds, expressed in basis points.
2. Interpret what this spread indicates about the market's perception of relative default risk between the two issuers.

Case 26: Static Spread versus the Simple Nominal Yield Spread

STAR MEDIA BROADCASTING LTD

Learning Objective: To compute a simple nominal yield spread between a corporate bond and a government benchmark, and to recognize its key limitation as a preview of the more rigorous Static Spread measure covered later.

Case Situation

An analyst comparing Star Media Broadcasting Ltd's 10-year corporate bond against a comparable 10-year government security observes that Star Media's bond yields 9.2%, while the government bond yields 7.1%. Before proceeding to the more rigorous spread measures used later in credit analysis, the analyst wants to first compute this simple nominal yield spread and understand precisely what assumption it relies upon that more advanced spread measures are specifically designed to correct.

Required

1. Compute the simple nominal yield spread between the two bonds, in basis points.
2. Identify the key simplifying assumption this spread measure relies upon.

Session 4: Duration

Cases 27-38

This session builds Macaulay and Modified Duration from first principles, applies them to price-change estimation and portfolio construction, and closes with duration-based immunization — the single most tested skill set in fixed-income coursework.

Case 27: Macaulay Duration — Full Worked Computation

ASHOKA ENGINEERING WORKS LTD

Learning Objective: To compute Macaulay Duration from first principles as the present-value-weighted average time to receipt of a bond's cash flows, and to interpret the resulting figure relative to maturity.

Case Situation

Ashoka Engineering Works Ltd has issued a 4-year bond with a face value of ₹1,000 and a 10% annual coupon, currently priced at par because its required yield also stands at 10%. A fixed-income trainee has been asked to compute the bond's Macaulay Duration from scratch, building the full present-value-weighted cash flow table, rather than relying on a shortcut formula, in order to build genuine intuition for what the duration figure actually represents.

Required

1. Compute the present value of each of the bond's four annual cash flows.
2. Compute the time-weighted present value of each cash flow.
3. Compute Macaulay Duration as the ratio of total time-weighted present value to total price.

Case 28: Macaulay Duration of a Zero-Coupon Bond

NOVA BIOTECH LTD

Learning Objective: To confirm, by direct computation, that a zero-coupon bond's Macaulay Duration always equals its time to maturity exactly.

Case Situation

Nova Biotech Ltd has issued a 7-year zero-coupon bond with a face value of ₹1,000, paying no periodic interest and redeemable only at maturity. A finance instructor wants to use this bond as a teaching example to demonstrate, through direct calculation rather than assertion, why a zero-coupon bond represents the one special case where Macaulay Duration exactly equals the bond's stated maturity, the upper limit that duration can never exceed for any bond.

Required

1. Compute the bond's Macaulay Duration using the standard present-value-weighting approach.
2. Confirm that this figure equals the bond's 7-year maturity exactly, and explain why.

Case 29: Macaulay Duration — The Effect of Coupon Rate

ZENITH AUTO FINANCE LTD

Learning Objective: To isolate the effect of coupon rate on Macaulay Duration by comparing two bonds identical in every respect except their coupon rate.

Case Situation

Zenith Auto Finance Ltd has two bonds outstanding, both with 10-year maturities and both currently yielding 9% to maturity, differing only in their coupon rates: one carries a 5% annual coupon, and the other a 12% annual coupon. A fixed-income training program wants trainees to compute and compare the Macaulay Duration of each bond, isolating the pure effect of coupon rate while holding maturity and yield constant.

Required

1. Compute the Macaulay Duration of the 5%-coupon bond.
2. Compute the Macaulay Duration of the 12%-coupon bond.
3. State and explain the direction of the relationship between coupon rate and duration observed.

Case 30: Macaulay Duration — The Effect of Term to Maturity**RIVERSIDE PAPER MILLS LTD**

Learning Objective: To isolate the effect of term to maturity on Macaulay Duration by comparing two bonds identical in every respect except their remaining maturity.

Case Situation

Riverside Paper Mills Ltd has two bonds outstanding, both carrying an identical 8% annual coupon and both currently yielding 8% to maturity, differing only in remaining maturity: one has 5 years remaining, and the other 15 years remaining. A portfolio risk team wants to quantify, in concrete duration terms, how much additional interest rate sensitivity the longer bond carries purely as a function of its longer remaining life.

Required

1. Compute the Macaulay Duration of the 5-year bond.
2. Compute the Macaulay Duration of the 15-year bond.
3. State whether duration increases proportionally with maturity, and explain the pattern observed.

Case 31: Modified Duration from Macaulay Duration**SKYLINE REALTY & INFRASTRUCTURE LTD**

Learning Objective: To convert Macaulay Duration into Modified Duration, the measure directly usable for estimating a bond's percentage price change for a given change in yield.

Case Situation

An analyst has already computed the Macaulay Duration of a Skyline Realty & Infrastructure Ltd bond as 7.5 years, using annual coupon payments and a Yield to Maturity of 9%. Before this duration figure can be used to estimate the bond's likely price response to an upcoming interest rate change, the analyst needs to convert it into Modified Duration, the measure specifically designed for that purpose.

Required

1. Compute the bond's Modified Duration from its Macaulay Duration.
2. Explain why Modified Duration, rather than Macaulay Duration directly, is the appropriate measure for estimating price sensitivity.

Case 32: Estimating Price Change Using Modified Duration**METRO RETAIL VENTURES LTD**

Learning Objective: To apply Modified Duration to estimate a bond's approximate percentage and absolute price change for a given change in yield.

Case Situation

A Metro Retail Ventures Ltd bond currently trades at ₹1,000 and has a Modified Duration of 6.8 years, as previously computed by the portfolio's fixed-income analyst. The desk anticipates that market yields will rise by 75 basis points over the coming quarter, following recent commentary from the central bank. The portfolio manager has asked the analyst to estimate the bond's approximate price impact under this yield-change scenario before the quarterly risk report is finalized.

Required

1. Estimate the bond's approximate percentage price change for the anticipated 75-basis-point yield increase.
2. Convert this percentage estimate into an approximate absolute price change and resulting new price.

Case 33: Price Value of a Basis Point (PVBP)**PRIME CONSUMER ELECTRONICS LTD**

Learning Objective: To compute the Price Value of a Basis Point, translating Modified Duration into the absolute rupee price change a bond experiences for a one-basis-point change in yield.

Case Situation

A trading desk holding a large position in Prime Consumer Electronics Ltd bonds, currently priced at ₹1,000 with a Modified Duration of 8.2 years, needs a practical, position-sizing-friendly measure of interest rate risk for daily risk reporting. Rather than express sensitivity as a percentage price change for an arbitrary yield move, the desk's risk manager wants the exact rupee price change corresponding to the smallest standard unit of yield movement traders actually discuss: a single basis point.

Required

1. Compute the bond's Price Value of a Basis Point (PVBP), also called DV01.
2. Explain how this figure would be used to size a position's rupee interest rate risk.

Case 34: Portfolio Duration as a Weighted Average**CONTINENTAL FINANCE & LEASING LTD**

Learning Objective: To compute the Modified Duration of a bond portfolio as the market-value-weighted average of its individual holdings' durations.

Case Situation

Continental Finance & Leasing Ltd's treasury desk holds a fixed-income portfolio consisting of two positions: ₹40 lakh invested in a bond with a Modified Duration of 4 years, and ₹60 lakh invested in a second bond with a Modified Duration of 9 years. Before the quarterly asset-liability committee meeting, the treasury team needs to determine the portfolio's overall interest rate sensitivity as a single combined duration figure.

Required

1. Compute the market-value weight of each bond within the total portfolio.
2. Compute the portfolio's overall Modified Duration as the weighted average of the two holdings.

Case 35: Duration-Based Immunization — Matching Asset and Liability Duration

SURAKSHA HOUSING FINANCE LTD

Learning Objective: To construct a two-bond portfolio whose blended Modified Duration matches a specified target liability duration, the foundational technique of duration-based immunization.

Case Situation

Suraksha Housing Finance Ltd's pension trust has a future benefit obligation with an estimated Modified Duration of 7 years. To protect the funding position from interest rate risk, the trust's investment committee wants to construct a bond portfolio using two available instruments, a shorter bond with a Modified Duration of 3 years, and a longer bond with a Modified Duration of 12 years, allocated in whatever proportion causes the blended portfolio duration to exactly match the 7-year liability duration.

Required

1. Determine the allocation weights between the two bonds required to achieve a blended portfolio duration of exactly 7 years.
2. Explain the principle of immunization that this allocation is designed to achieve.

Case 36: Repositioning a Bond Portfolio for an Expected Rate Decline

GOLDEN HARVEST AGRO LTD

Learning Objective: To reallocate a bond portfolio's composition in order to increase its overall Modified Duration ahead of an anticipated fall in interest rates.

Case Situation

Golden Harvest Agro Ltd's treasury desk currently holds a bond portfolio allocated 60% to a shorter bond with a Modified Duration of 3 years and 40% to a longer bond with a Modified Duration of 9 years. Following a change in the desk's macroeconomic outlook, the head of treasury now expects interest rates to decline meaningfully over the coming year and wants to reposition the portfolio to capture more upside from this anticipated move, by shifting the allocation to 30% in the shorter bond and 70% in the longer bond.

Required

1. Compute the portfolio's current blended Modified Duration under the existing 60:40 allocation.

2. Compute the portfolio's new blended Modified Duration under the proposed 30:70 allocation.
3. Explain why this reallocation is an appropriate response to an expected decline in interest rates.

Case 37: Repositioning a Bond Portfolio for an Expected Rate Increase

VAYU WIND ENERGY LTD

Learning Objective: To reallocate a bond portfolio's composition in order to reduce its overall Modified Duration ahead of an anticipated rise in interest rates.

Case Situation

Vayu Wind Energy Ltd's treasury desk currently holds a bond portfolio allocated equally, 50% each, between a shorter bond with a Modified Duration of 4 years and a longer bond with a Modified Duration of 10 years. Following commentary suggesting an imminent monetary policy tightening cycle, the head of treasury now expects interest rates to rise over the coming year and wants to reposition the portfolio defensively, shifting the allocation to 80% in the shorter bond and 20% in the longer bond to reduce exposure to the expected rate increase.

Required

1. Compute the portfolio's current blended Modified Duration under the existing 50:50 allocation.
2. Compute the portfolio's new blended Modified Duration under the proposed 80:20 allocation.
3. Explain why this reallocation is an appropriate defensive response to an expected rise in interest rates.

Case 38: The Effect of a Sinking Fund or Call Provision on Duration

SILVERLINE SHIPYARDS LTD

Learning Objective: To understand, through comparative computation, how a sinking fund or call provision shortens a bond's effective cash-flow schedule and thereby reduces its Macaulay Duration relative to an otherwise identical bullet-maturity bond.

Case Situation

Silverline Shipyards Ltd has issued a 10-year, 9% annual coupon bond priced at par, since its required yield also stands at 9%. An analyst wants to compare this bond's Macaulay Duration against what it would be if the bond instead carried a sinking fund provision requiring the issuer to retire the bond's principal early, effectively shortening the bond's average cash-flow schedule to resemble a 7-year instrument rather than a full 10-year bullet maturity bond.

Required

1. Compute the Macaulay Duration of the plain 10-year bullet-maturity bond.
2. Compute the Macaulay Duration of the bond under the sinking fund scenario, approximated as an effective 7-year cash-flow schedule.
3. State the direction and approximate size of the duration reduction caused by the sinking fund provision.

Session 5: Convexity & Embedded Options

Cases 39-49

This session adds the second-order refinement of convexity to the duration toolkit, then turns to callable and puttable bonds, where Effective Duration and Effective Convexity are required precisely because standard duration measures assume cash flows that these instruments do not guarantee.

Case 39: Measuring Convexity — Actual versus Duration-Predicted Price Change

EVEREST MINING & MINERALS LTD

Learning Objective: To compare a bond's actual price change against the change predicted by Modified Duration alone, isolating the effect of convexity for a materially large yield change.

Case Situation

Everest Mining & Minerals Ltd has a 15-year, 9% annual coupon bond, currently priced at par with a required yield of 9% and a computed Modified Duration of 8.06 years. Market yields have since risen by a full 100 basis points, to 10%. A risk analyst wants to compare the bond's actual, fully recomputed price at the new yield against the price decline that Modified Duration alone would have predicted, to quantify how much convexity cushions the actual outcome.

Required

1. Estimate the bond's percentage price decline using Modified Duration alone.
2. Compute the bond's actual percentage price decline by fully repricing it at the new 10% yield.
3. Quantify the difference between the two figures and attribute it to convexity.

Case 40: Combining Duration and Convexity for a Price Estimate

PARADIP PORT HOLDINGS LTD

Learning Objective: To combine a duration-based estimate with a convexity adjustment to produce a more accurate second-order approximation of a bond's price change for a large yield move.

Case Situation

A Paradip Port Holdings Ltd bond has a Modified Duration of 7.5 years and a Convexity measure of 85. Market analysts are forecasting a sharp 200-basis-point rise in yields over the coming year in response to anticipated monetary tightening. Given the size of this anticipated move, a fixed-income analyst has been asked to produce a more accurate price-change estimate than Modified Duration alone would provide, by incorporating the bond's convexity into the calculation.

Required

1. Compute the duration-only estimate of percentage price change for the anticipated 200-basis-point yield increase.
2. Compute the convexity adjustment term and add it to the duration-only estimate.

3. State the combined, second-order estimate of the bond's percentage price change.

Case 41: The Value of a Callable Bond

SUNRISE BANK OF INDIA LTD

Learning Objective: To value a callable bond as the value of an otherwise identical non-callable bond, reduced by the value of the embedded call option retained by the issuer.

Case Situation

Sunrise Bank of India Ltd has issued a 10-year bond with an 11% annual coupon and a face value of ₹1,000, embedded with a call provision that permits early redemption by the issuer. An otherwise identical non-callable version of this bond, at the current 9% required yield, would be worth ₹1,128.35. An option-pricing model has separately estimated the value of the embedded call option, given current interest rate volatility and the bond's specific call schedule, at ₹45. A bond analyst has been asked to determine the fair value of the actual callable bond.

Required

1. State the relationship linking a callable bond's value to the value of an equivalent non-callable bond and the embedded call option.
2. Compute the fair value of the callable bond using the given figures.

Case 42: Callable Bond Price Response to a Rate Decline

CRESTVIEW FINANCIAL SERVICES LTD

Learning Objective: To observe, through comparative computation, how a callable bond's price appreciation is capped near its call price as interest rates decline substantially, in contrast to the larger appreciation of an otherwise identical non-callable bond.

Case Situation

Crestview Financial Services Ltd has an 8-year, 10% annual coupon callable bond, with a call price of ₹1,050. At the current 9% required yield, an equivalent non-callable version of this bond is worth ₹1,055.35. Market yields then fall sharply to 6%, a substantial 300-basis-point decline. An equivalent non-callable bond would be worth ₹1,248.39 at this new lower yield. A portfolio manager wants to understand how the callable bond's actual price would behave differently under this same rate decline.

Required

1. Compute how much the equivalent non-callable bond's price would rise given the 300-basis-point yield decline.
2. Explain why the callable bond's actual price rise would be much smaller, and estimate its approximate resulting price.

Case 43: The Value of a Puttable Bond

UNITY GENERAL INSURANCE LTD

Learning Objective: To value a puttable bond as the value of an otherwise identical non-puttable bond, increased by the value of the embedded put option held by the investor.

Case Situation

Unity General Insurance Ltd has issued a 10-year bond with a 7% annual coupon and a face value of ₹1,000, embedded with a put provision that permits the investor to sell the bond back to the issuer at a predetermined price before maturity. An otherwise identical non-puttable version of this bond, at the current 9% required yield, would be worth ₹871.65. An option-pricing model has separately estimated the value of the embedded put option, given current interest rate volatility and the bond's specific put schedule, at ₹38. A bond analyst has been asked to determine the fair value of the actual puttable bond.

Required

1. State the relationship linking a puttable bond's value to the value of an equivalent non-puttable bond and the embedded put option.
2. Compute the fair value of the puttable bond using the given figures.

Case 44: Puttable Bond Price Response to a Rate Increase

LAKEVIEW RESORTS & HOTELS LTD

Learning Objective: To observe, through comparative computation, how a puttable bond's price decline is cushioned near its put price as interest rates rise substantially, in contrast to the larger decline of an otherwise identical non-puttable bond.

Case Situation

Lakeview Resorts & Hotels Ltd has an 8-year, 8% annual coupon puttable bond, with a put price of ₹950. At the current 9% required yield, an equivalent non-puttable version of this bond is worth ₹944.65. Market yields then rise sharply to 13%, a substantial 400-basis-point increase. An equivalent non-puttable bond would fall to ₹760.06 at this new higher yield. A portfolio manager wants to understand how the puttable bond's actual price would behave differently under this same rate increase.

Required

1. Compute how much the equivalent non-puttable bond's price would fall given the 400-basis-point yield increase.
2. Explain why the puttable bond's actual price decline would be much smaller, and estimate its approximate resulting price.

Case 45: Effective Duration for a Bond with an Embedded Option

FALCON AVIATION SERVICES LTD

Learning Objective: To compute Effective Duration using three prices generated by an option-pricing model, the correct duration measure for bonds whose cash flows change with interest rates.

Case Situation

Falcon Aviation Services Ltd has a callable bond currently priced at ₹1,000. Using an option-pricing model that accounts for the embedded call feature, an analyst has estimated the bond's price would rise to ₹1,035 if yields fell by 50 basis points, and would fall to ₹970 if yields instead rose by 50 basis points. Because Macaulay and Modified Duration both assume fixed cash flows, an assumption this callable bond violates, the analyst needs to compute Effective Duration instead, using these three option-model-derived prices.

Required

1. Compute the bond's Effective Duration using the three given prices.
2. Explain why this approach is necessary instead of the standard Modified Duration formula for this bond.

Case 46: Effective Convexity for a Bond with an Embedded Option

FALCON AVIATION SERVICES LTD

Learning Objective: To compute Effective Convexity using the same three option-model-derived prices used for Effective Duration, completing the second-order interest rate risk picture for a bond with an embedded option.

Case Situation

Continuing the same analysis of Falcon Aviation Services Ltd's callable bond — currently priced at ₹1,000, rising to ₹1,035 if yields fall 50 basis points, and falling to ₹970 if yields rise 50 basis points — the analyst now wants to complete the risk picture by computing Effective Convexity from the same three prices, in order to check whether this callable bond exhibits the negative convexity behavior expected of instruments with issuer-held call options.

Required

1. Compute the bond's Effective Convexity using the three given prices.
2. State whether the result is consistent with negative convexity, and explain what a negative Effective Convexity value would mean.

Case 47: Empirical Duration from Observed Price and Yield Changes

ORBIT TELECOMMUNICATIONS LTD

Learning Objective: To compute Empirical Duration directly from an actually observed price change and yield change in the market, without reference to any theoretical cash-flow or option-pricing model.

Case Situation

An Orbit Telecommunications Ltd bond traded at ₹1,000 last month. Following a broad market move in benchmark yields of 50 basis points, the bond's observed market price fell to ₹965. A quantitative analyst wants to estimate this bond's Empirical Duration directly from this observed price and yield behavior, as a real-world cross-check against the bond's theoretically modeled duration figure.

Required

1. Compute the bond's percentage price change from the observed market data.

2. Compute the bond's Empirical Duration using this observed percentage price change and the observed yield change.

Case 48: Effective Duration Across the Interest Rate Cycle — Callable Bond

SUNRISE BANK OF INDIA LTD

Learning Objective: To observe how a callable bond's Effective Duration itself changes as interest rate levels move, contracting sharply as yields fall and expanding toward the duration of an equivalent non-callable bond as yields rise.

Case Situation

Sunrise Bank of India Ltd's callable bond has had its Effective Duration estimated by an option-pricing model at two different points in the interest rate cycle. When market yields were relatively low, close to making the embedded call economically attractive to the issuer, Effective Duration was estimated at just 2.1 years. When market yields later rose substantially, making a call highly unlikely, Effective Duration was re-estimated at 7.8 years. A risk manager wants this pattern explained clearly before presenting it to the bank's asset-liability committee.

Required

1. State which of the two Effective Duration figures corresponds to the low-yield environment and which to the high-yield environment.
2. Explain why Effective Duration changes so substantially between the two interest rate environments for this callable bond.

Case 49: Effective Duration Across the Interest Rate Cycle — Puttable Bond

LAKEVIEW RESORTS & HOTELS LTD

Learning Objective: To observe how a puttable bond's Effective Duration itself changes as interest rate levels move, contracting sharply as yields rise and expanding toward the duration of an equivalent non-puttable bond as yields fall.

Case Situation

Lakeview Resorts & Hotels Ltd's puttable bond has had its Effective Duration estimated by an option-pricing model at two different points in the interest rate cycle. When market yields were relatively low, making the embedded put unattractive to investors, Effective Duration was estimated at 7.6 years, close to that of an equivalent non-puttable bond. When market yields later rose substantially, making exercising the put highly attractive, Effective Duration was re-estimated at just 2.4 years. A risk manager wants this pattern explained clearly before presenting it to the company's asset-liability committee.

Required

1. State which of the two Effective Duration figures corresponds to the low-yield environment and which to the high-yield environment.
2. Explain why Effective Duration changes so substantially between the two interest rate environments for this puttable bond, and contrast the direction of this pattern with the callable bond case.

Session 6: Advanced Spread Analysis & Integrative Cases

Cases 50-57

This closing session introduces the Static (Zero-Volatility) Spread and the Option-Adjusted Spread, then brings every tool in this casebook together in four fully integrative capstone cases spanning yield, duration, convexity, and spread analysis in combination.

Case 50: Computing the Static (Zero-Volatility) Spread

SAHYADRI ENERGY HOLDINGS LTD

Learning Objective: To compute the Static (Zero-Volatility) Spread of a corporate bond by finding the constant spread that, added to every point on the government spot rate curve, correctly reprices the bond to its observed market value.

Case Situation

Horizon Capital Bank Ltd has issued a 3-year, 9.5% annual coupon bond with a face value of ₹1,000, currently trading at ₹990. The prevailing government spot rate curve stands at 6.0% for one year, 6.5% for two years, and 7.0% for three years. Rather than rely on a simple nominal yield spread, a credit analyst wants to compute the more precise Static (Zero-Volatility) Spread, which correctly accounts for the shape of the entire government spot curve rather than a single benchmark point.

Required

1. State the equation defining the Static Spread and identify what it solves for.
2. Solve for the Static Spread that reprices Horizon Capital's bond to its observed ₹990 market price.

Case 51: Comparing Two Bonds Using Static Spread

KONKAN MARINE ENERGY LTD

Learning Objective: To use Static Spread as a relative-value tool for identifying which of two comparable bonds offers superior compensation for credit and liquidity risk, after correctly adjusting for the shape of the government spot curve.

Case Situation

A relative-value desk is comparing two similarly-rated corporate bonds from issuers in the same sector as Konkan Marine Energy Ltd, both with comparable maturities. Bond A has a computed Static Spread of 145 basis points over the government spot curve, while Bond B has a computed Static Spread of 110 basis points over the same curve. Both bonds carry comparable credit ratings according to published agency assessments. The desk has been asked to determine which bond represents the more attractive relative-value opportunity on a spread basis.

Required

1. Identify which bond offers the wider Static Spread.

2. Explain what this spread difference suggests about relative value between the two comparably rated bonds.

Case 52: Why Static Spread Overstates Value for a Callable Bond

SUNRISE BANK OF INDIA LTD

Learning Objective: To recognize why the Static Spread measure, despite correctly adjusting for the government curve's shape, still overstates the true credit-and-liquidity compensation offered by a bond with an embedded call option.

Case Situation

A credit analyst has computed a Static Spread of 310 basis points for a callable bond issued by Sunrise Bank of India Ltd, compared with a Static Spread of only 180 basis points for an otherwise similar, but non-callable, bond from a comparably rated peer institution. Before recommending the callable bond as offering superior relative value purely on the basis of its wider spread, the analyst wants to understand what portion of this 130-basis-point gap is genuine additional compensation, and what portion is simply an artifact of the embedded call option.

Required

1. Explain why the Static Spread calculation itself does not distinguish between compensation for credit risk and compensation for the embedded call option.
2. State what adjustment is needed before the two bonds' spreads can be fairly compared.

Case 53: Option-Adjusted Spread — A Comparative Case

SUNRISE BANK OF INDIA LTD

Learning Objective: To understand, at a conceptual and comparative level, how the Option-Adjusted Spread isolates a callable bond's true credit-and-liquidity compensation from the value attributable to its embedded call option.

Case Situation

Continuing the analysis of Sunrise Bank of India Ltd's callable bond, whose Static Spread was computed at 310 basis points against a non-callable peer's 180 basis points, an option-pricing desk has now run an interest rate simulation model to estimate the bond's Option-Adjusted Spread (OAS) directly. The model, which explicitly incorporates the probability of the bond being called under a wide range of simulated future interest rate paths, produces an OAS estimate of 195 basis points. The analyst wants to interpret this figure and finally resolve the earlier comparison against the non-callable peer bond.

Required

1. State how the 195-basis-point OAS estimate relates to the bond's 310-basis-point Static Spread.
2. Compare this OAS figure against the non-callable peer bond's 180-basis-point spread and draw a relative-value conclusion.

Case 54: Integrative Case: Yield, Duration, and Convexity for a Single Bond

BHILAI STEEL ROLLING MILLS LTD

Learning Objective: To apply Yield to Maturity, Macaulay Duration, Modified Duration, and Convexity together on a single bond, integrating the full toolkit developed across this casebook into one comprehensive analysis.

Case Situation

Bhilai Steel Rolling Mills Ltd has a 10-year bond outstanding with a face value of ₹1,000 and a 9% annual coupon, currently trading at a required yield of 8.5%. A senior analyst is preparing a comprehensive one-page risk summary for this single bond ahead of a client presentation, and wants every major analytical tool covered in this casebook — price, duration, and convexity — brought together into one integrated view, culminating in a combined price-change estimate for a plausible future yield move.

Required

1. Compute the bond's current fair value at the given 8.5% required yield.
2. Compute the bond's Macaulay Duration, Modified Duration, and Convexity.
3. Estimate the bond's percentage price change for a 100-basis-point rise in yield, combining the duration and convexity effects.

Case 55: Integrative Case: Comparing Two Bonds for a Portfolio Decision**KONKAN ENGINEERING WORKS LTD**

Learning Objective: To compare two bonds of differing coupon structure but identical maturity and yield across price, duration, and risk characteristics, in order to make a reasoned portfolio allocation decision.

Case Situation

A portfolio manager at a fund holding Konkan Engineering Works Ltd exposure is choosing between two 7-year bonds carrying the same 8.5% required yield: Bond A, a low-coupon bond paying 6% annually, and Bond B, a high-coupon bond paying 11% annually. The manager has a moderately cautious near-term interest rate outlook, expecting yields to rise modestly over the next year, and needs a side-by-side comparison of both bonds' prices and durations before deciding which better suits the fund's current risk positioning.

Required

1. Compute the price and Macaulay Duration of Bond A.
2. Compute the price and Macaulay Duration of Bond B.
3. Recommend which bond better suits a moderately cautious, rising-rate outlook, and justify the recommendation.

Case 56: Integrative Case: Immunizing a Pension Liability**SURAKSHA HOUSING FINANCE LTD**

Learning Objective: To apply duration-matching immunization to a realistic pension liability, determining the correct portfolio allocation across two available bonds and confirming the resulting protection against interest rate risk.

Case Situation

Suraksha Housing Finance Ltd's employee pension trust has a future benefit obligation with a present value of ₹500 lakh and an estimated Modified Duration of 8 years. To immunize this liability, the trust's investment committee has access to two bonds: a shorter instrument with a Modified Duration of 4 years, and a longer instrument with a Modified Duration of 11 years. The committee needs both the correct rupee allocation across the two bonds and a brief confirmation of how this allocation protects the pension trust's funding position.

Required

1. Determine the portfolio weights in each bond required to match the liability's 8-year duration.
2. Convert these weights into rupee allocations across the ₹500 lakh portfolio.
3. Explain how this allocation protects the trust's funding position against interest rate movements.

Case 57: Integrative Case: Full Analysis of a Callable Bond

SUNRISE BANK OF INDIA LTD

Learning Objective: To conduct a complete callable-bond analysis in one integrated case, combining Yield to Call, Effective Duration, negative convexity, and Option-Adjusted Spread into the single comprehensive framework a fixed-income analyst would actually use in practice.

Case Situation

As a capstone exercise, a senior fixed-income analyst has been asked to prepare a complete risk-and-return profile for Sunrise Bank of India Ltd's callable bond, last examined earlier in this casebook. The bond has an 11% annual coupon, a face value of ₹1,000, trades at ₹1,080, and is callable at ₹1,050 after 4 years. An option-pricing model has separately estimated the bond's Effective Duration at 4.2 years, versus 7.1 years for an equivalent non-callable bond, and its Option-Adjusted Spread at 195 basis points. The analyst must integrate all of this into one coherent summary.

Required

1. Compute the bond's Yield to Call, assuming redemption at the first call date.
2. Compare the given Effective Duration against the equivalent non-callable bond's duration, and interpret the gap.
3. State how the Option-Adjusted Spread should be used relative to the bond's other yield and spread measures in this summary.